SENNA ECL Nepal — Complete User Guide

A plain-language guide to Expected Credit Loss (ECL) under NFRS 9 / IFRS 9, the Nepal Rastra Bank (NRB) guidelines, and how this system calculates it.

Written for risk officers, finance teams, auditors, and anyone new to ECL.

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1. What is Expected Credit Loss (ECL)?

When a bank lends money, some borrowers may not repay. **ECL is the amount the bank expects to lose** on its loans, calculated *before* the loss actually happens.

Under the old rules, banks only recognised losses after a borrower defaulted (incurred loss). Under **NFRS 9 (Nepal Financial Reporting Standards 9)**, banks must recognise losses **from day one** — forward-looking, probability-weighted, and discounted.

```
ECL = PD × LGD × EAD
```

- **PD** = Probability of Default (will the borrower fail to pay?)
- **LGD** = Loss Given Default (how much will we lose if they default?)
- **EAD** = Exposure at Default (how much money is at risk?)

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2. Regulatory framework in Nepal

2.1 Who must comply

All banks and financial institutions (BFIs) licensed under the Bank and Financial Institution Act, 2073 that prepare financial statements under NFRS.

2.2 Key NRB documents

Document	Issued	Key content
NFRS 9 — ECL Related Guidelines, 2024	Jan/May 2024	Staging, PD/LGD/EAD, floors, governance,
First Amendment (2082/83)	Jul 2025	Calendar-month delinquency backstops, PO
Guidance Note on Interest Income Recogni	Jul 2025	Quarterly, previous-quarter stage
Unified Directives (Class D)	Ongoing	Prudential provision, NPL classification

2.3 Applicability timeline

- ICAN carve-out deferred full NFRS 9 implementation until FY 2080/81.
- From FY 2081/82, all BFIs must use the ECL impairment model.
- CET1 transition: 80% → 60% → 40% → 20% → 0% add-back over 5 years.

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3. How ECL is calculated (step by step)

Step 1 — Stage classification

Every loan is placed in one of three stages based on credit risk:

Stage	Condition	ECL horizon
Stage 1	Performing, no significant increase in c	12-month ECL
Stage 2	SICR since origination (but not yet defa	Lifetime ECL

Stage 3	Credit-impaired / defaulted	Lifetime ECL (with recovery assessment)

Step 2 — NRB calendar-month backstops (2082/83 amendment)

Days past due (DPD)	Stage backstop
Up to 30 days (1 month)	Stage 1
31–90 days (1–3 months)	Stage 2
91+ days (more than 3 months)	Stage 3

DPD is a **backstop**, not the whole policy. Qualitative triggers (watchlist,

forbearance, covenant breach, bankruptcy, etc.) can also move a loan to Stage 2/3.

Step 3 — Calculate PD (Probability of Default)

Infinity's PD model (per SAR Advisors methodology) works in 6 steps:

1. **Define default** and build the default history (observed default rates by segment and year).
2. **Estimate TTC PD** (through-the-cycle, long-run default rate) = average of annual default rates over 5+ years.
3. **Calibrate Vasicek** asset correlation (ρ) and credit cycle index (Z).
4. **Link Z to macroeconomic variables** (GDP, inflation, credit growth, sector indicators) via regression.
5. **Project scenarios** (base/best/worst) and convert TTC→PIT PD for each year.
6. **Build term structure** (12-month and lifetime) and apply the 2.5% NRB floor.

Vasicek formula (standard form):

```
PIT_PD = Φ( (Φ⁻¹(TTC_PD) + √ρ · Z) / √(1 − ρ) )
```

- Φ = standard normal CDF
- TTC_PD = through-the-cycle default rate
- ρ = asset correlation
- Z = credit cycle index (positive = downturn/adverse conditions → higher PIT PD)

Note: at Z=0 the PIT PD is NOT equal to TTC PD. TTC PD is the expectation over Z~N(0,1).

- **12-month PD**: likelihood of default within 12 months.
- **Lifetime PD**: likelihood of default over the remaining life of the loan.
- PD must use historical data + forward-looking macro information (no proxies).
- **NRB floor**: minimum 2.5% PD for all exposures.

Step 4 — Calculate LGD (Loss Given Default)

- LGD = 1 − Recovery Rate
- First use **observed historical recovery** experience.
- If no history: use collateral value with prudential haircut (25% amended).
- Collateral not realized within 5 years of default cannot be used.
- **NRB floor**: minimum 45% LGD if no reliable history (board approval required).
- GoN-guaranteed exposures: minimum 0% LGD (20% for FX).
- Subordinated claims: minimum 75% LGD.

Step 5 — Calculate EAD (Exposure at Default)

- EAD = Outstanding balance + Undrawn limit × Credit Conversion Factor (CCF).
- Includes accrued interest and fees.
- For revolving facilities, model expected future drawdowns.

Step 6 — Discount to present value

- ECL is discounted using the **Effective Interest Rate (EIR)**.
- Formula: ECL = Σ [PD(t) × (1 − cumulative PD) × LGD(t) × EAD(t) × DF(t)]

Step 7 — Scenario weighting

Run the calculation under 3 macro scenarios:

Scenario	Typical weight	Adjustments
Optimistic	20%	PD × 0.7, LGD × 0.9
Base	50%	PD × 1.0, LGD × 1.0
Pessimistic	30%	PD × 1.4, LGD × 1.15

Final ECL = weighted average of all three.

Step 8 — Apply overlays (PMA)

Management overlays (post-model adjustments) for factors not captured by the model:

- Sector stress, pandemic, natural disaster, regulatory changes.

- Each overlay requires documented reason and approval.

Step 9 — Compare with prudential provision

NRB requires: **Provision = max(Calculated ECL, NRB Prudential Provision)**.

The "higher-of" rule ensures ECL never falls below the prudential floor.

Step 10 — Interest income recognition (NRB 2025)

Stage	Method	Basis
Stage 1 & 2	Effective Interest Rate (EIR)	Gross carrying amount
Stage 3	Cash basis	Actual cash received

- Uses the **previous quarter's stage** for the current quarter's interest recognition.
- Requires quarterly account-level evidence from FY 2083/84.

4. Data requirements — what details are needed

4.1 Minimum required (V0.1 pilot)

Field	Source	Description
account_id	CBS	Unique loan account number
customer_id	CBS	Customer identifier
outstanding	CBS	Current outstanding principal
origination_date	CBS	Loan disbursement date
maturity_date	CBS	Loan maturity date
dpd	CBS	Days past due
classification	CBS	NRB classification (Pass/Substandard/Dou

4.2 Recommended (V0.2+) — per SAR Advisors data request

Field	Source	Why needed
undrawn_limit	CBS	For EAD calculation
interest_rate / EIR	CBS	Discounting
collateral_value + type + valuation date	CBS/Valuation	LGD calculation
cash collateral, guarantees	CBS	Recovery sources
sector	CBS	Segmentation
branch / geographical location	CBS	Branch-level reporting
watchlist flag	Risk system	SICR trigger
forbearance flag	Risk system	SICR trigger
repayment schedule	CBS	EAD term structure
transactions (with reversals)	CBS	Cash flow validation
macro indicators (GDP, inflation, credit	NRB/IMF/ADB	Forward-looking scenarios
historical stage/default/recovery	CBS	PD/LGD calibration
defaulted/closed accounts (date of first	CBS	LGD workout
off-balance sheet (undrawn limits, commi	CBS	EAD/CCF
credit risk ratings (origination + curre	Risk system	SICR / segmentation
NRB provisioning by category, regulatory	Finance	Reconciliation
investment/placement details	Treasury	Other exposures

4.3 Period covered

- At least **5 years** of loan-level history (NRB requirement for PD).
- **7 years** recommended where the system holds it (more Vasicek observations).
- For LGD: all defaulted accounts closed or recovered.

4.3 NRB 2.2 report fields (Infinity's format)

The system reads Infinity's NRB 2.2 report directly:

- `AC_NO` → account_id
- `CUSTOMER_CODE` → customer_id
- `F_OUTSTANDING_PRINCIPAL` → outstanding
- `F_APPROVED_LIMIT` → undrawn_limit (derived)
- `INT_RATE` → interest_rate
- `OVERDUE_DAYS` → dpd
- `CLASSIFICATION` → NRB classification (1=Pass, 1.1=Watchlist, 3=Substandard, 4=Doubtful, 5=Loss)
- `CAT_NAME` → sector
- `BR_NAME` → branch
- `F_TYPE_OF_CREDIT` → product

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5. POCI (Purchased or Originated Credit Impaired)

A new loan disbursed in the current period to an account that was **Stage 3 at the previous reporting date**

is treated as POCI. This requires:

- Previous-quarter stage history.
- Old-to-new account linkage (top-up/renewal detection).
- Lifetime ECL from origination (not from SICR date).

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6. NRB floors and overrides (versioned rule pack)

Parameter	Floor	When applied
PD	2.5% minimum	All exposures
LGD (no history)	45% minimum	Board approval required
Collateral haircut	25%	Amended wording
GoN-guaranteed LGD	0% minimum	Government-guaranteed
FX GoN-guaranteed LGD	20% minimum	Foreign currency
Subordinated claims LGD	75% minimum	Subordinated debt

Transparency rule: the system always shows raw model output, floor applied, final parameter,

and reason. It never silently replaces model estimates.

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7. Reports generated by the system

#	Report	Format
1	ECL Summary by Stage	Excel/CSV/JSON
2	Stage Migration Matrix	Excel/CSV/JSON
3	Account-level ECL Detail (with triggers)	Excel/CSV/JSON
4	PD/LGD/EAD by Segment	Excel/CSV/JSON
5	Scenario Analysis & Sensitivity	Excel/CSV/JSON
6	Provision Comparison (ECL vs Prudential)	Excel/CSV/JSON
7	IFRS 7 Disclosure Tables	Excel/CSV/JSON
8	Vintage Analysis	Excel/CSV/JSON
9	Branch/Sector Summaries	Excel/CSV/JSON
10	Interest Income Reconciliation	Excel/CSV/JSON
11	Data Quality & Validation Report	Excel/CSV/JSON
12	Audit Trail / Calculation Manifest	Excel/CSV/JSON

13	Movement Analysis (ECL Bridge)	Excel/CSV/JSON

7b. SICR indicators (per NRB 2024 / SAR Advisors)

Quantitative (test each against institution data):

- More than 30 days past due
- Absolute lifetime PD of 5% or more
- Relative lifetime PD increase of 100% or more since origination
- Risk rating downgraded by 2 notches, or to non-investment grade

Qualitative:

- Deterioration in borrower's cash flows or financial position
- Expectation of forbearance/restructuring due to financial difficulty
- Modification of terms resulting in restructuring/rescheduling
- Deterioration of prospects for the borrower's sector
- Borrowers affected by adverse macroeconomic conditions (supportable forecasts)
- Weaker credit quality indicators in individual monitoring
- Management decision to strengthen collateral/covenants due to rising risk

The 30 days past due indicator is a **backstop that cannot be overridden** by a

favorable model result.

7c. Segmentation (collective vs individual)

- **Collective**: retail and small-ticket loans, business loans below the large-exposure threshold — grouped by shared risk characteristics.
- **Individual**: large exposures, Stage 3 exposures, exposures with borrower-specific information.
- Segment tests: homogeneity (default rates alike), data sufficiency (enough defaults), stability (consistent across years).
- Restructured/rescheduled loans tracked as a separate flag.
- Segmentation reviewed at least annually and on new products / material portfolio shifts.

7d. Probation and stage transfers

- Stage 2 → Stage 1: only with evidence of significant reduction in credit risk (Board-approved criteria).
- Stage 3 exit: minimum 180 days probation.
- Restructured Stage 3 exit: minimum 24 months probation.
- Restructured exposures: no move back to 12-month ECL without sufficient evidence of improved credit risk.

7e. Back-testing and model monitoring

- Compare predicted defaults, recoveries, and losses with actual outcomes at least annually.
- Recommend model adjustments where differences are material.
- Independent validation by a reviewer independent of the model-building team.
- Internal Audit independently reviews data integrity, policy compliance, and follow-up of findings.

7f. NRB quarterly reporting and transition

- **NRB Annexes 1–4**: submitted within 15 days of quarter end.
- **Day 1 impact & CET1 transition**: add-back over 5 years (85% / 65% / 50% / 30% / 15%).
- **NFRS 7 disclosures**: stage-wise carrying amounts, loss allowance reconciliation, stage migration, write-offs and recoveries.
- **Regulatory reserve**: institution recognizes the higher of ECL and NRB Unified Directives provisions; reconciled quarterly.

7g. Board-approved ECL policy (minimum contents)

Roles and responsibilities; control procedures for model development/validation;

criteria for individual and collective assessment (including large exposure definition);

SICR indicators; staging criteria (qualitative + quantitative); criteria for moving to a lower stage;

sources of forward-looking information (macro factors).

8. How to use the system (step by step)

1. **Prepare data**: Download the vendor template (`templates/SENNA_ECL_Vendor_Data_Template.xlsx`) or use Infinity's NRB 2.2 report directly.
2. **Open the frontend**: `streamlit run apps/frontend/app.py`
3. **Select data source**: Bundled sample / Full Infinity data / Upload file.
4. **Set parameters**: Tenant name, as-of date, institution ID.
5. **Click "Run ECL"**: The system runs 4 steps:
   - Ingest & validate (schema, types, control totals)
   - ECL calculation (staging → PD/LGD/EAD → scenarios)
   - Generate reports (Excel/CSV/JSON)
   - NRB provision comparison
6. **Review**: KPIs, stage summary, account table, provision comparison.
7. **Download**: Excel, CSV, JSON reports.

8. **Audit**: Check the pipeline trace and run history.

9. Glossary

Term	Meaning
ECL	Expected Credit Loss — probability-weigh
PD	Probability of Default
LGD	Loss Given Default
EAD	Exposure at Default
SICR	Significant Increase in Credit Risk
POCI	Purchased or Originated Credit Impaired
DPD	Days Past Due
EIR	Effective Interest Rate
PMA	Post-Model Adjustment (overlay)
NRB	Nepal Rastra Bank
NFRS	Nepal Financial Reporting Standards
IFRS	International Financial Reporting Standa
CET1	Common Equity Tier 1 capital
BS	Bikram Sambat (Nepal calendar)
AD	Anno Domini (Gregorian calendar)

10. Important notes

- All outputs are **PRELIMINARY/RESEARCH** until the release gates are met.
- The system never silently replaces model estimates with floors — it shows raw, floor, final, and reason.
- No production customer data should be committed to Git.
- Independent model validation is required before statutory use.